

Proof. Let

$$\mathbf{a} = (a_1, \dots, a_n), \quad \mathbf{b} = (b_1, \dots, b_n) \in \mathbb{R}^n.$$

Define the Euclidean inner product and norm

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=1}^n a_i b_i, \quad \|\mathbf{a}\|^2 = \langle \mathbf{a}, \mathbf{a} \rangle = \sum_{i=1}^n a_i^2, \quad \|\mathbf{b}\|^2 = \langle \mathbf{b}, \mathbf{b} \rangle = \sum_{i=1}^n b_i^2.$$

Degenerate cases. If $\mathbf{a} = \mathbf{0}$ or $\mathbf{b} = \mathbf{0}$ then $\langle \mathbf{a}, \mathbf{b} \rangle = 0$ and also $\|\mathbf{a}\|^2 = 0$ or $\|\mathbf{b}\|^2 = 0$. Hence

$$(\langle \mathbf{a}, \mathbf{b} \rangle)^2 = 0 = \|\mathbf{a}\|^2 \|\mathbf{b}\|^2,$$

so the inequality holds with equality. Henceforth assume $\mathbf{a} \neq \mathbf{0}$ and $\mathbf{b} \neq \mathbf{0}$; consequently $\|\mathbf{a}\|^2 > 0$ and $\|\mathbf{b}\|^2 > 0$.

A non-negative quadratic. For a real parameter t define

$$Q(t) = \sum_{i=1}^n (a_i - t b_i)^2.$$

Each summand is a square, therefore $Q(t) \geq 0$ for every $t \in \mathbb{R}$. Expanding,

$$\begin{aligned} Q(t) &= \sum_{i=1}^n (a_i^2 - 2t a_i b_i + t^2 b_i^2) \\ &= \|\mathbf{a}\|^2 - 2t \langle \mathbf{a}, \mathbf{b} \rangle + t^2 \|\mathbf{b}\|^2. \end{aligned}$$

Thus

$$Q(t) = \|\mathbf{b}\|^2 t^2 - 2\langle \mathbf{a}, \mathbf{b} \rangle t + \|\mathbf{a}\|^2,$$

a quadratic polynomial with positive leading coefficient $\|\mathbf{b}\|^2 > 0$.

Discriminant condition. A quadratic $At^2 + Bt + C$ with $A > 0$ satisfies $At^2 + Bt + C \geq 0$ for all real t iff its discriminant is non-positive:

$$B^2 - 4AC \leq 0.$$

Applying this to the coefficients

$$A = \|\mathbf{b}\|^2, \quad B = -2\langle \mathbf{a}, \mathbf{b} \rangle, \quad C = \|\mathbf{a}\|^2,$$

we obtain

$$\Delta = (-2\langle \mathbf{a}, \mathbf{b} \rangle)^2 - 4\|\mathbf{b}\|^2 \|\mathbf{a}\|^2 \leq 0.$$

Since

$$(-2\langle \mathbf{a}, \mathbf{b} \rangle)^2 - 4\|\mathbf{b}\|^2\|\mathbf{a}\|^2 = 4(\langle \mathbf{a}, \mathbf{b} \rangle^2 - \|\mathbf{a}\|^2\|\mathbf{b}\|^2),$$

dividing the inequality $\Delta \leq 0$ by the positive constant 4 yields

$$\langle \mathbf{a}, \mathbf{b} \rangle^2 \leq \|\mathbf{a}\|^2 \|\mathbf{b}\|^2.$$

Writing the inner products and norms in component form gives the **Cauchy–Schwarz inequality**

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right).$$

Equality condition. Equality holds precisely when $\Delta = 0$, i.e. when the quadratic has a double root $t_0 = \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\|\mathbf{b}\|^2}$. Since $Q(t_0) = 0$ and each term $(a_i - t_0 b_i)^2 \geq 0$, we must have

$$a_i = t_0 b_i \quad (i = 1, \dots, n).$$

Thus the vectors are linearly dependent; there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ for every i (the trivial case $\lambda = 0$ corresponds to one of the vectors being the zero vector).

Hence equality in the Cauchy–Schwarz inequality holds **iff** the two sequences are proportional.

□